

IPL DATA ANALYSIS

Mini Project Report

Python Analysis, Visualizations and Interactive Power BI Dashboard

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Project	IPL Performance Analytics
Dataset	IPL cleaned delivery-level dataset
Dashboard	IPL Performance Analysis Dashboard

Prepared for final mini-project submission.

1. Project Objective

The objective is to analyze IPL match and player-related data and convert raw records into meaningful sports insights. The project studies team and player performance, match outcomes, batting, bowling, venues, toss decisions and season-wise trends. Python is used for cleaning, analysis and visualizations, while Power BI is used for an interactive dashboard.

2. Dataset Description

The cleaned dataset is delivery-level data, so match information repeats for individual deliveries. For match-level questions such as winners, venues, toss decisions and Player of the Match, one record per Match_ID is used. Delivery-level rows are retained for batting, bowling, runs and wickets analysis.

Metric	Result
Total Matches	120
Total Runs	52,857
Total Wickets	1,386
Seasons	2015-2024
Average Runs per Match	440.48

3. Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI Desktop, Power Query and DAX.

4. Data Cleaning & Preprocessing

The dataset was checked for data types, missing values, duplicate records and inconsistent categorical values. Exact duplicate delivery records were removed, text values were trimmed, historical franchise names were standardized, selected missing values were handled, numeric columns were converted appropriately, and Match_Date was converted to date format. Derived fields were also created for toss-match outcomes and boundaries.

5. Data Analysis Performed

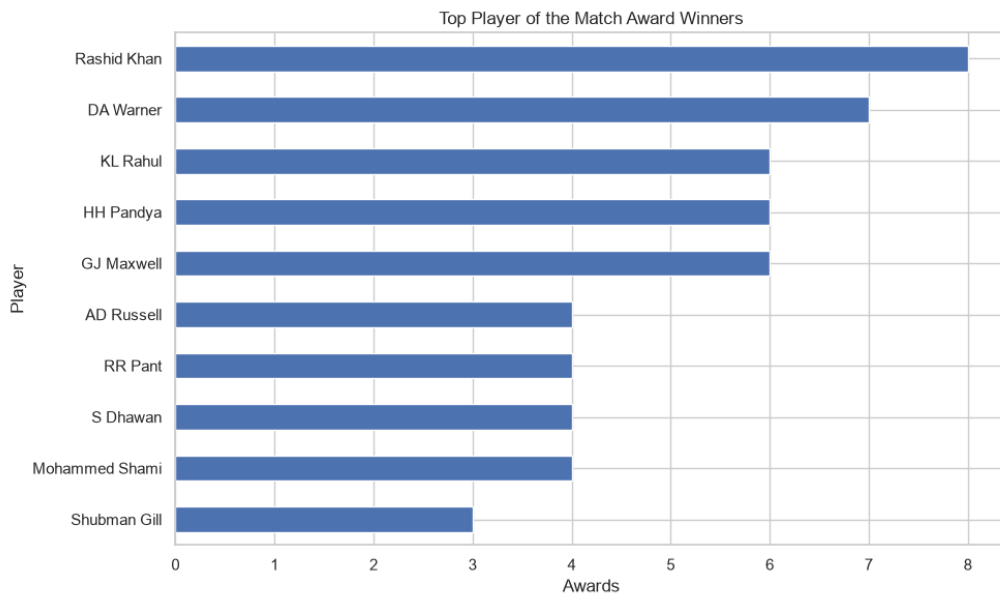
The analysis includes matches by season, team wins, Player of the Match awards, venue frequency, toss decisions, toss-to-match-win percentage, winning margins, top batsmen, top bowlers, season-wise runs and wickets, average runs, team-wise averages, team performance and correlations.

Analysis	Result
Highest team wins	Sunrisers Hyderabad - 17 wins
Highest run scorer	KL Rahul - 2,270 runs
Highest wicket taker	Mohammed Shami - 113 wickets
Most Player of the Match awards	Rashid Khan - 8 awards
Most-used venue	Eden Gardens - 19 matches
Most common toss decision	Field - 77 matches
Toss winner also won match	50.83%
Highest-run season	2023 - 5,401 runs
Highest team average runs	Lucknow Super Giants - 223.96

6. Python Visualizations

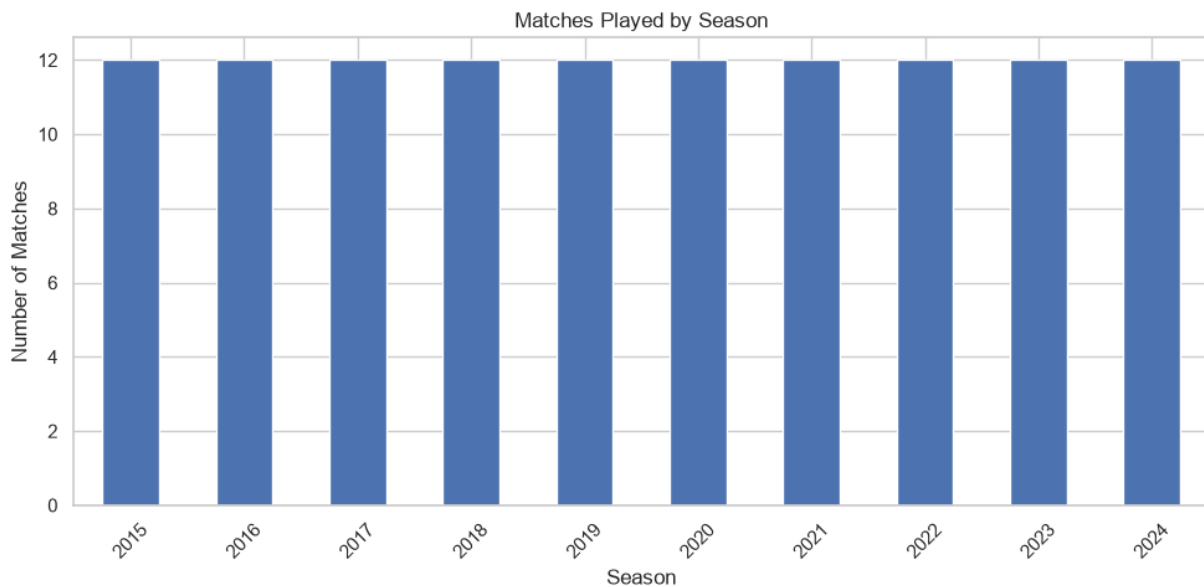
The following charts are the outputs generated by the Python analysis file.

Figure 1. Matches Played by Season



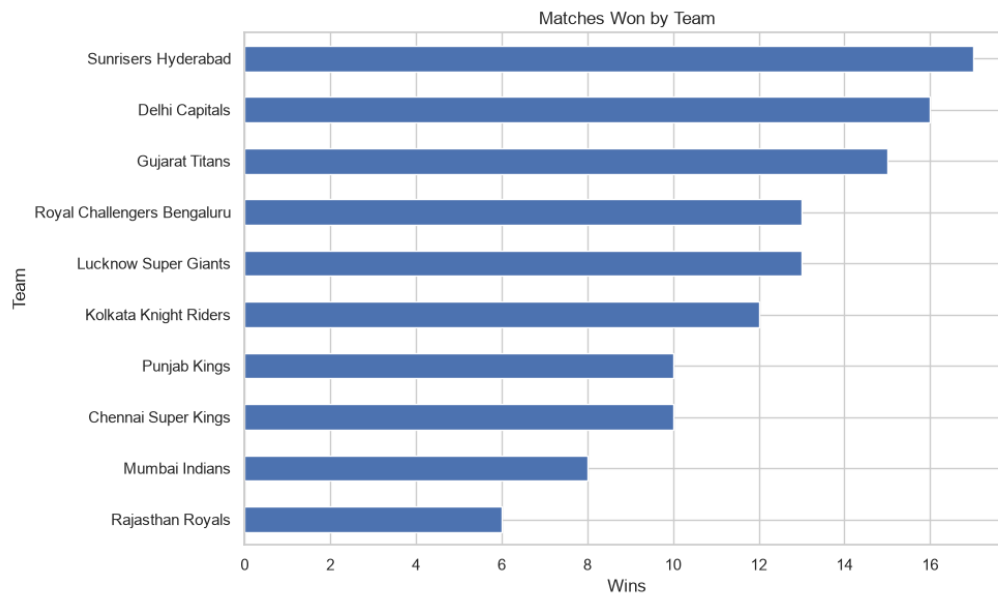
Each season from 2015 to 2024 contains 12 matches in the analyzed dataset.

Figure 2. Matches Won by Team



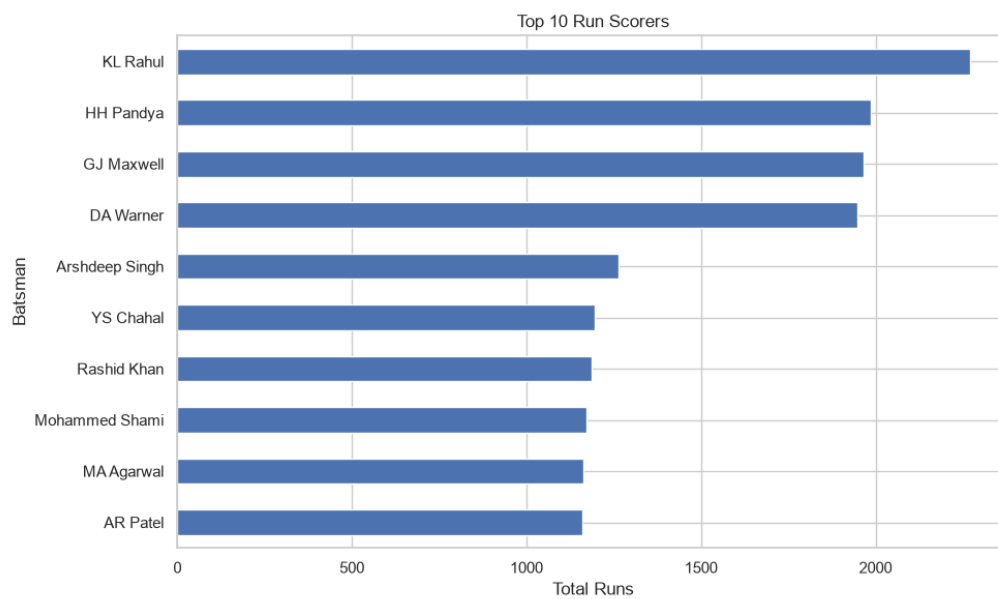
Sunrisers Hyderabad leads the dataset with 17 wins.

Figure 3. Top 10 Run Scorers



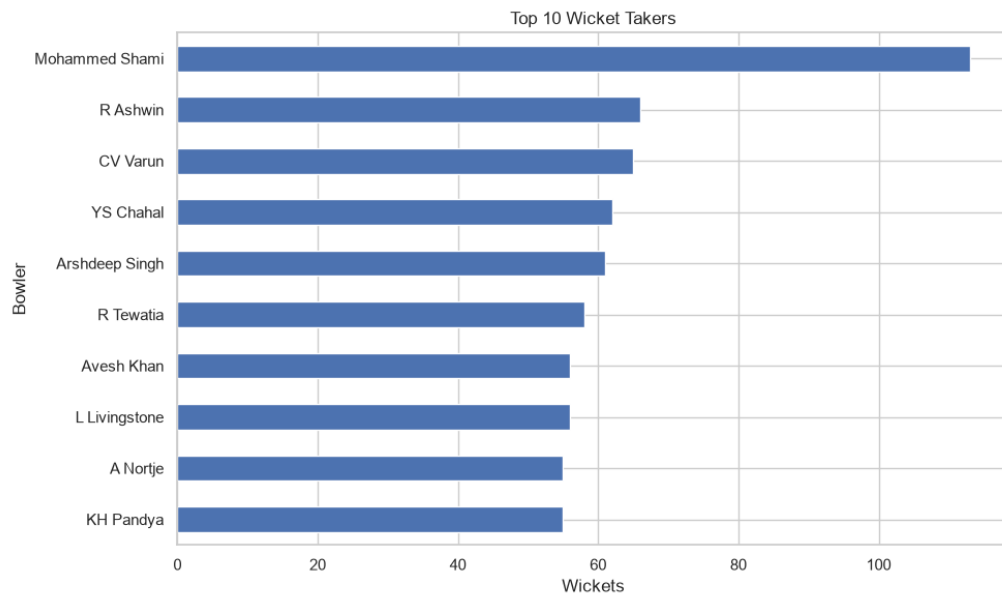
KL Rahul is the leading run scorer with 2,270 runs.

Figure 4. Top 10 Wicket Takers



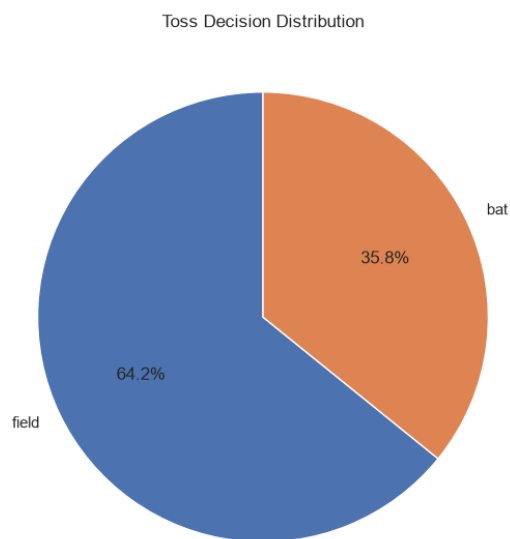
Mohammed Shami is the leading wicket taker with 113 wickets.

Figure 5. Player of the Match Award Winners



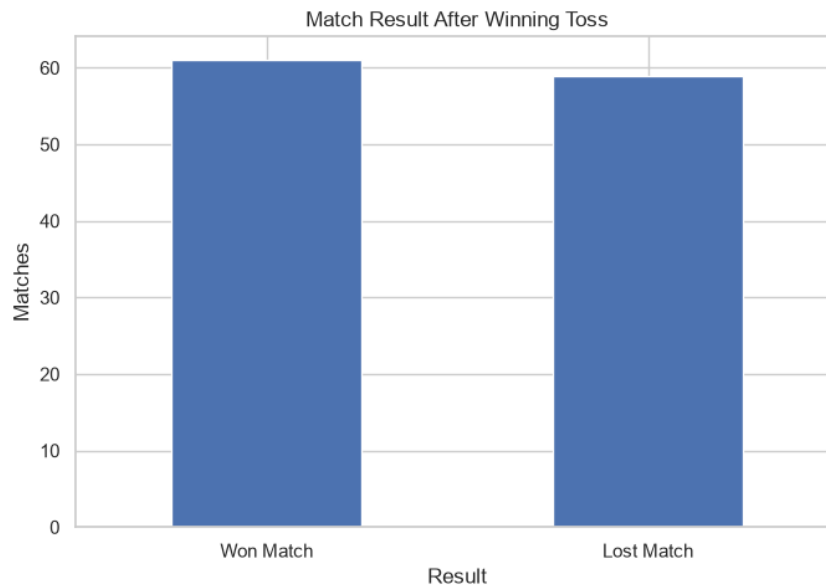
Rashid Khan has the most Player of the Match awards with 8.

Figure 6. Toss Decision Distribution



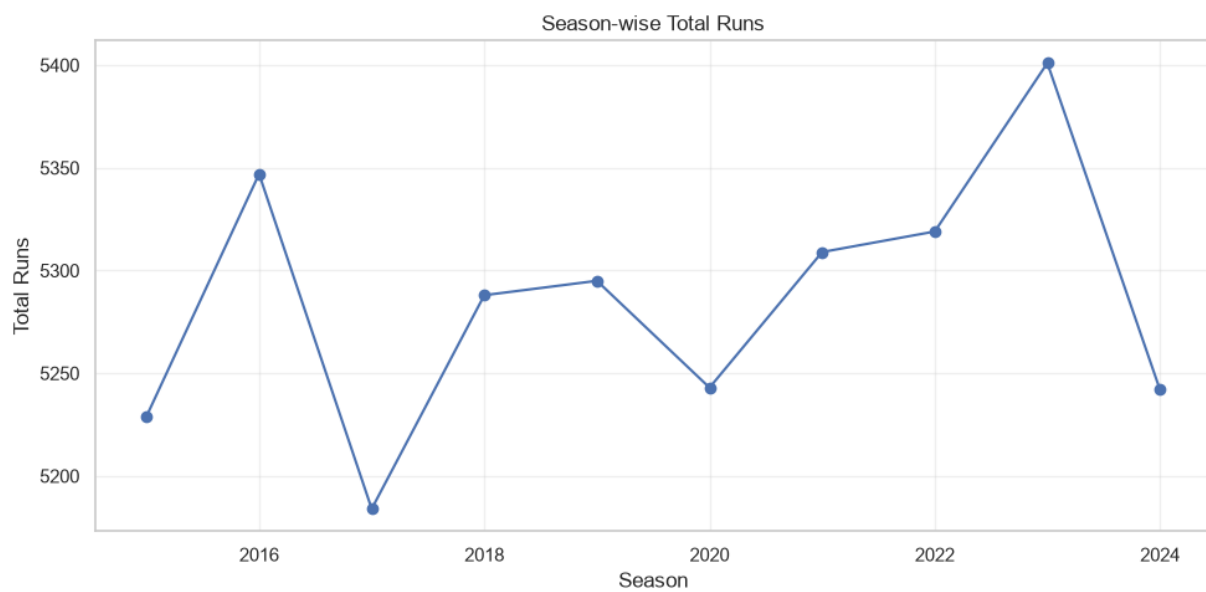
Fielding is the preferred toss decision, selected in approximately 64.2% of matches.

Figure 7. Match Result After Winning Toss



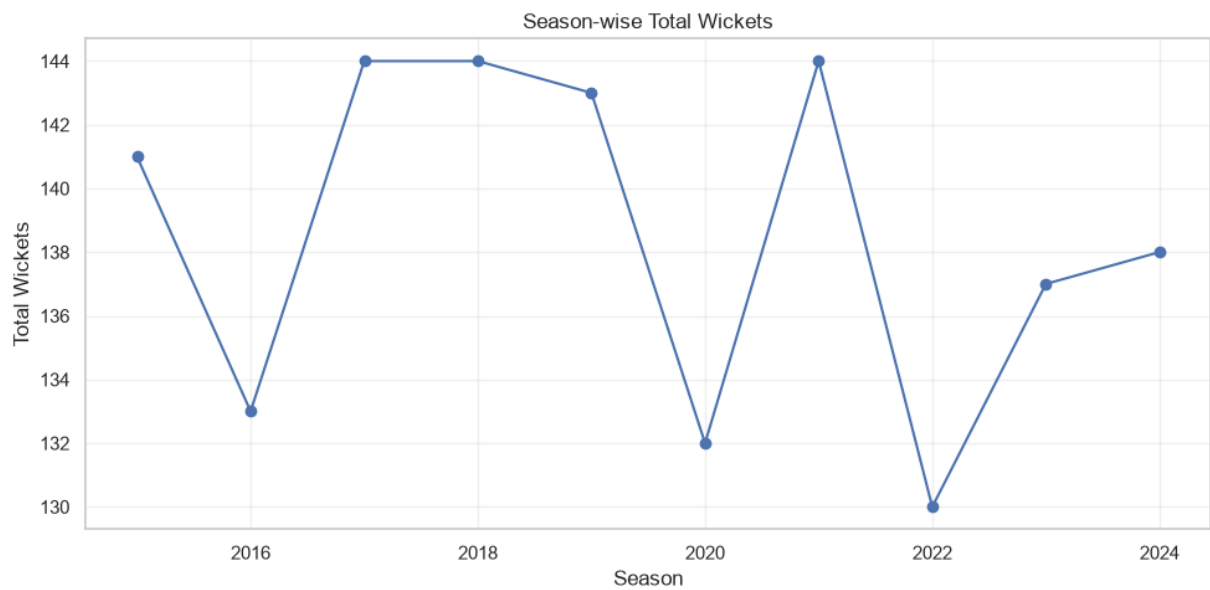
Toss winners won 61 matches and lost 59, showing only a small difference.

Figure 8. Season-wise Total Runs



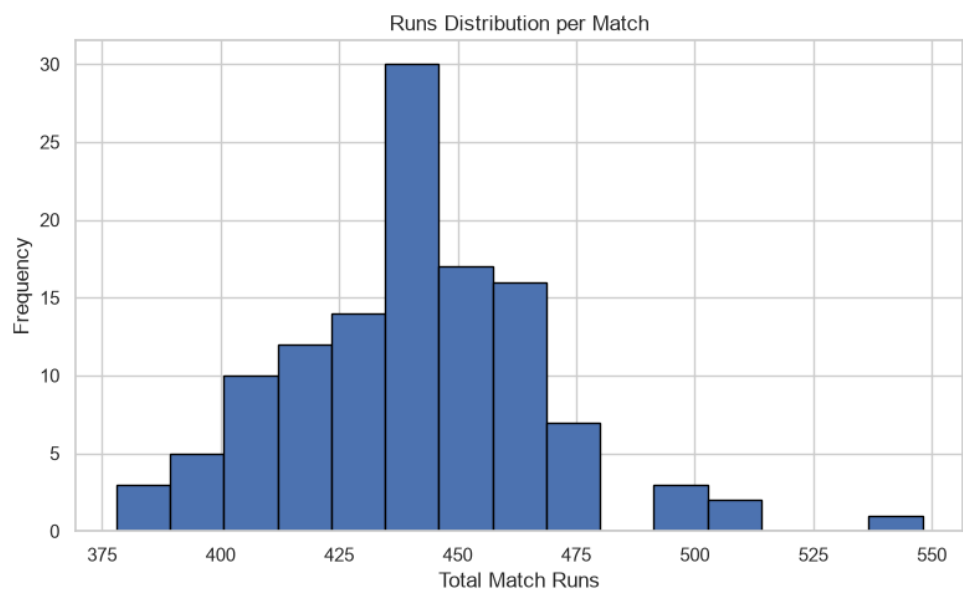
The highest run total occurs in 2023 with 5,401 runs.

Figure 9. Season-wise Total Wickets



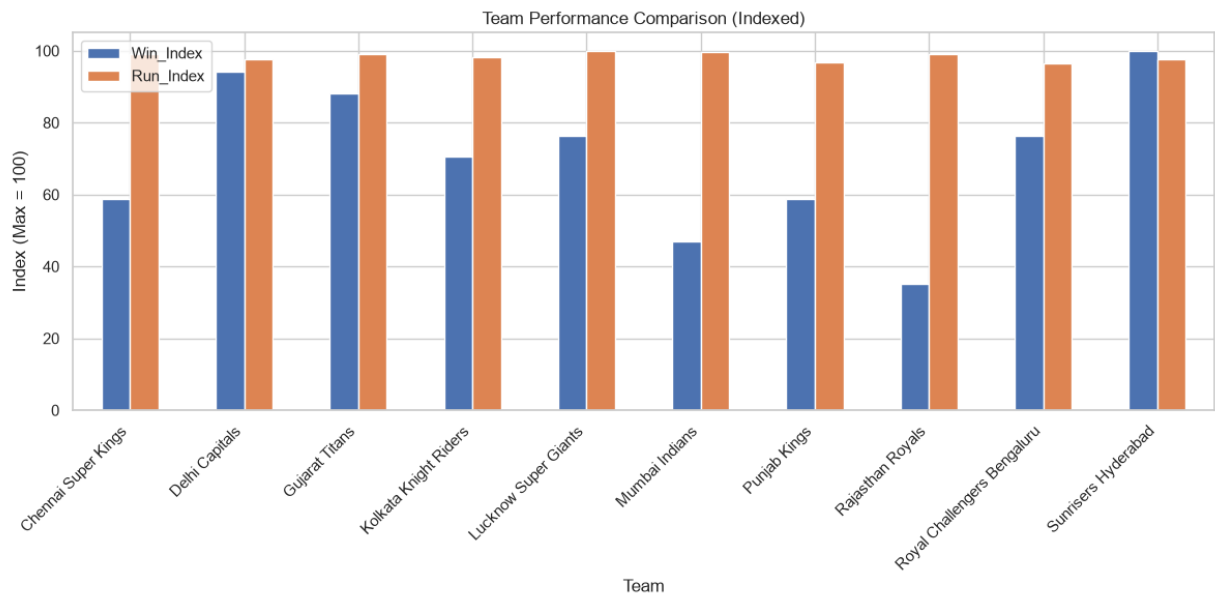
Season-wise wicket totals fluctuate within a relatively narrow range in this dataset.

Figure 10. Runs Distribution per Match



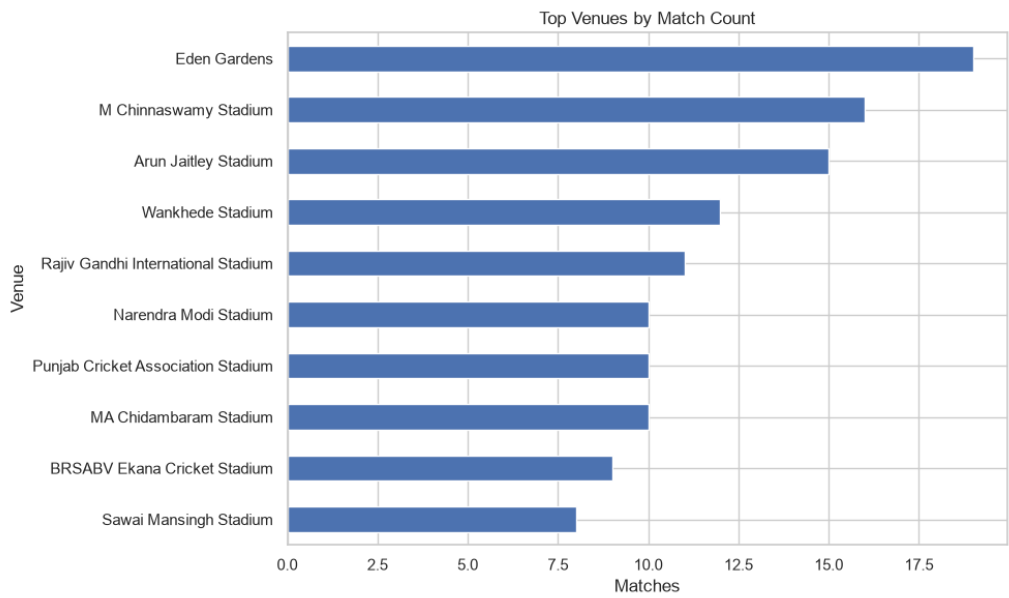
Most match totals are concentrated around the low-to-mid 400s, with a small number of higher-scoring matches.

Figure 11. Team Performance Comparison (Indexed)



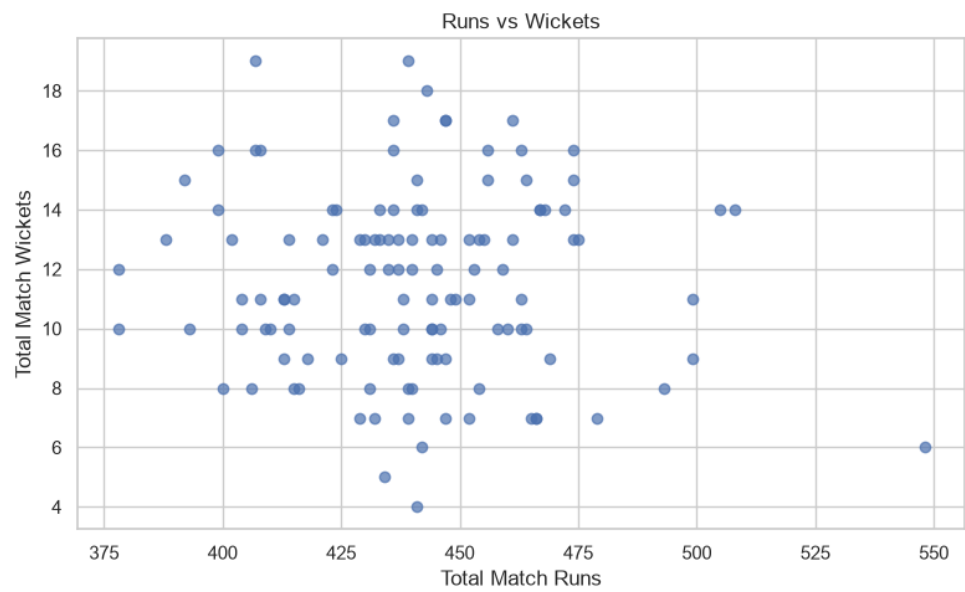
The indexed chart compares team win performance with average-run performance on a common 0-100 scale.

Figure 12. Top Venues by Match Count



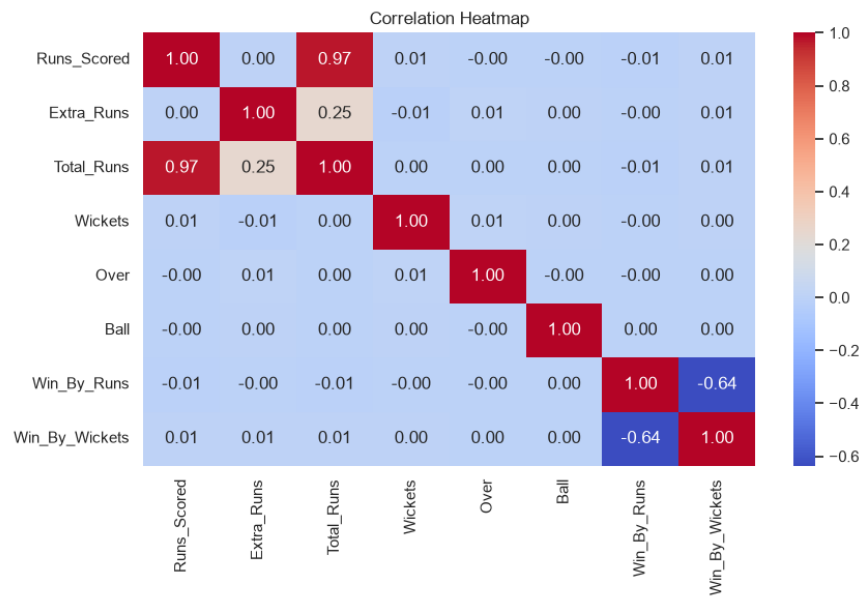
Eden Gardens is the most frequently used venue with 19 matches.

Figure 13. Runs vs Wickets



The scatter plot shows no strong simple linear relationship between total match runs and total wickets.

Figure 14. Correlation Heatmap



Runs_Scored and Total_Runs show a very strong positive correlation; most other displayed relationships are weak.

7. Power BI Dashboard

The final one-page Power BI dashboard combines KPI cards with team, season, player, venue and toss analysis. The dashboard is interactive through slicers and visual interactions.

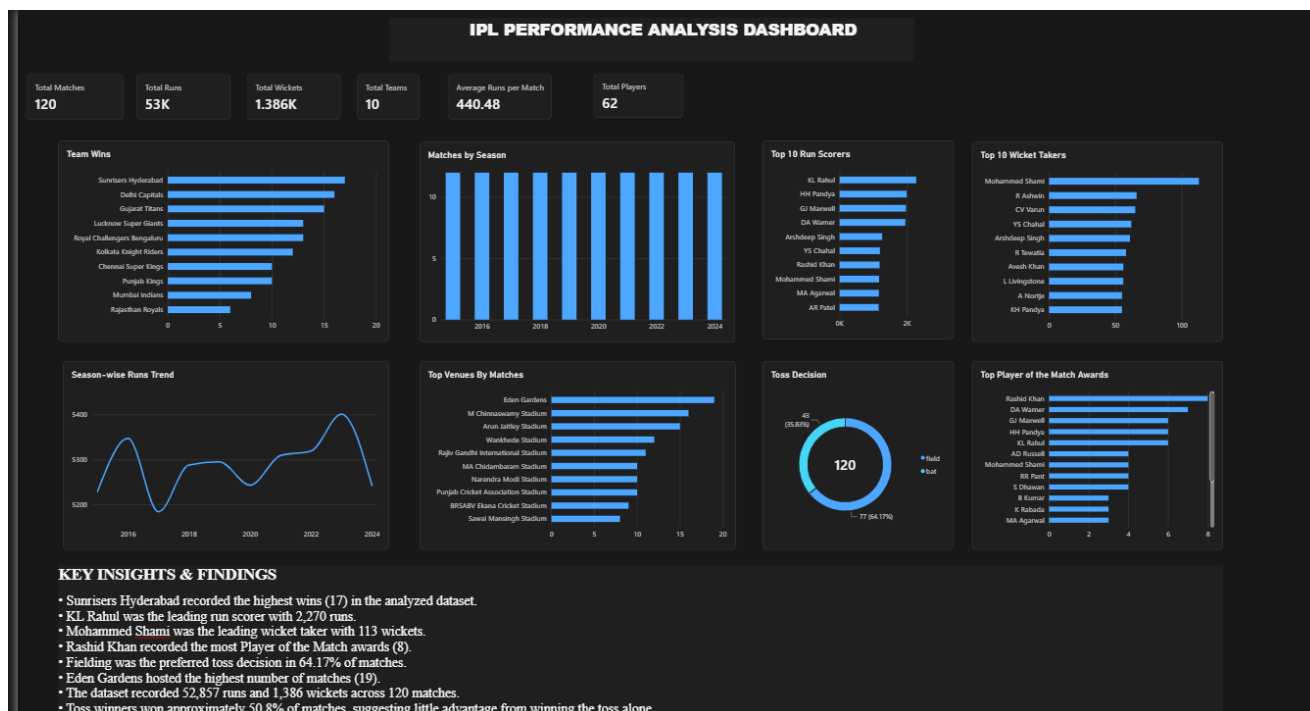


Figure 15. Final IPL Performance Analysis Dashboard

The dashboard summarizes 120 matches, approximately 53K runs, 1,386 wickets, 10 teams, 62 displayed player count and an average of 440.48 runs per match, together with the major analytical findings.

8. Business Insights - Questions & Answers

1. Which team has the highest number of wins?

Sunrisers Hyderabad has the highest number of wins with 17 victories.

2. Which season had the highest number of matches?

All seasons from 2015 to 2024 are tied at 12 matches each in this dataset.

3. Which player has received the highest Player of the Match awards?

Rashid Khan leads with 8 Player of the Match awards.

4. Who is the highest run scorer?

KL Rahul is the highest run scorer with 2,270 runs.

5. Who is the highest wicket taker?

Mohammed Shami is the highest wicket taker with 113 wickets.

6. Which venue hosted the highest number of matches?

Eden Gardens hosted the most matches with 19.

7. What is the most common toss decision?

Fielding is the most common decision: 77 matches (64.17%).

8. Does winning the toss appear to influence match results?

Toss winners also won the match in 50.83% of matches. This is close to 50%, so the dataset suggests little advantage from winning the toss alone.

9. Which season recorded the highest number of runs?

The 2023 season recorded the highest total with 5,401 runs.

10. Which team has the highest average runs?

Lucknow Super Giants has the highest team-wise average at approximately 223.96 runs per team-match.

11. What are the major performance trends?

Sunrisers Hyderabad leads team wins; KL Rahul, Mohammed Shami and Rashid Khan lead the main player measures. Run totals are fairly stable across seasons with a 2023 peak. Fielding is strongly preferred after winning the toss, but toss winners win only about half of matches.

12. Write 5-10 key analytical insights.

The key analytical insights are summarized in the next section.

9. Key Analytical Insights & Findings

1. Sunrisers Hyderabad recorded the highest number of wins with 17 victories.
2. KL Rahul was the leading run scorer with 2,270 runs.
3. Mohammed Shami was the leading wicket taker with 113 wickets.
4. Rashid Khan recorded the most Player of the Match awards with 8.
5. Fielding was selected in 77 of 120 matches (64.17%).
6. Eden Gardens hosted the highest number of matches with 19.
7. The 2023 season recorded the highest run total with 5,401 runs.
8. Toss winners also won approximately 50.83% of matches, suggesting little advantage from the toss alone.
9. Every season in the dataset contains 12 matches, so no single season leads by match count.
10. Lucknow Super Giants recorded the highest team-wise average runs at approximately 223.96.

10. Conclusion

This project demonstrates an end-to-end IPL data analysis workflow. Delivery-level data was cleaned and standardized in Python, analyzed with Pandas and NumPy, visualized through Matplotlib and Seaborn, and presented in an interactive Power BI dashboard. The analysis identifies leading teams and players, venue usage, toss behavior, season-wise trends and relationships between major match variables.